NSCC

Final Project

Ubuntu Install

Jeong Eun Jang (w0451032)

OSYS 1000-701

Paul Street

April 16th, 2021

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1. INTRODUCTION

Ubuntu is one of the most popular and widely used Linux distribution. Before starting the final project of the Operating Systems course, I had many options of choice Linux distribution. I was always curious about Ubuntu after I started my Linux with Linux Mint. So, I chose Ubuntu 18.04 which has long-term support(LTS) and has been chosen by many Linux users. Ubuntu uses GNOME(GNU Network Object Model Environment) desktop environment since the 18.04 LTS version released, with file manager Nautilus as the default installation and delivery application, Firefox as the Web browser, Libre Office as the document program.

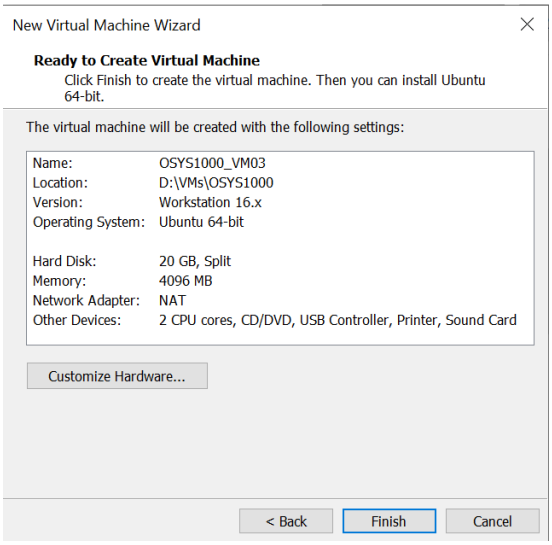
2. TASK 1: UBUNTU 18.04.5 LTS

2.1. Objectives for this task are as follows:

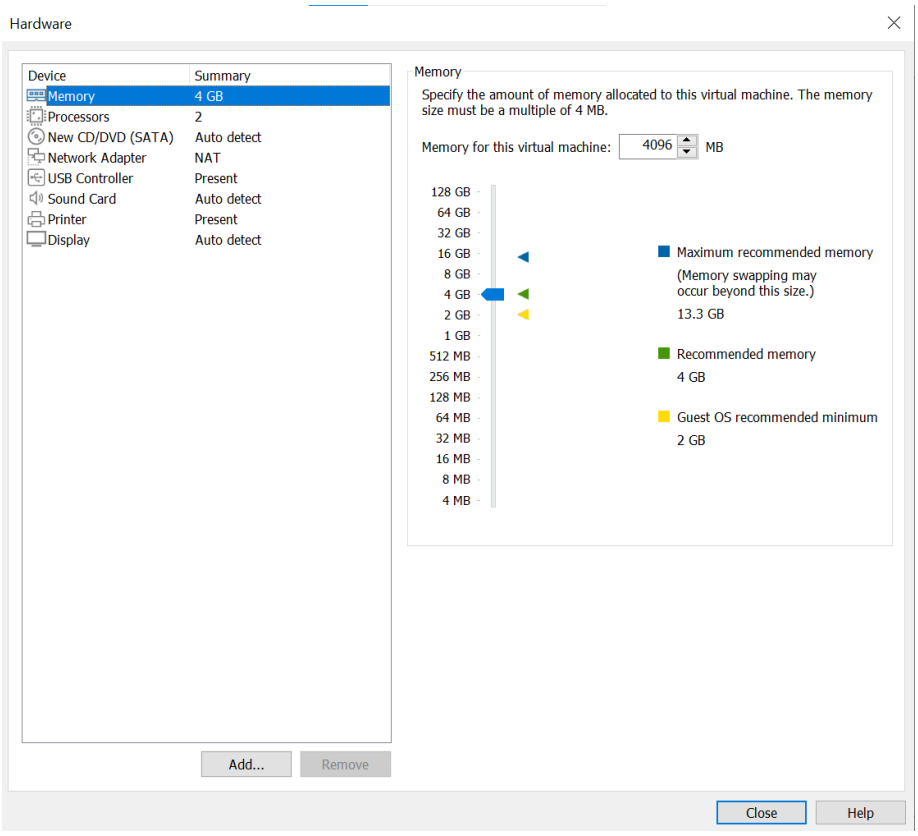
: Create a new Virtual Machine with Linux Ubuntu 18.04.5 LTS 64-bit and set an environment to start it.

2.2. Create a 'New' Virtual Machine with Ubuntu 18.04.5 LTS.

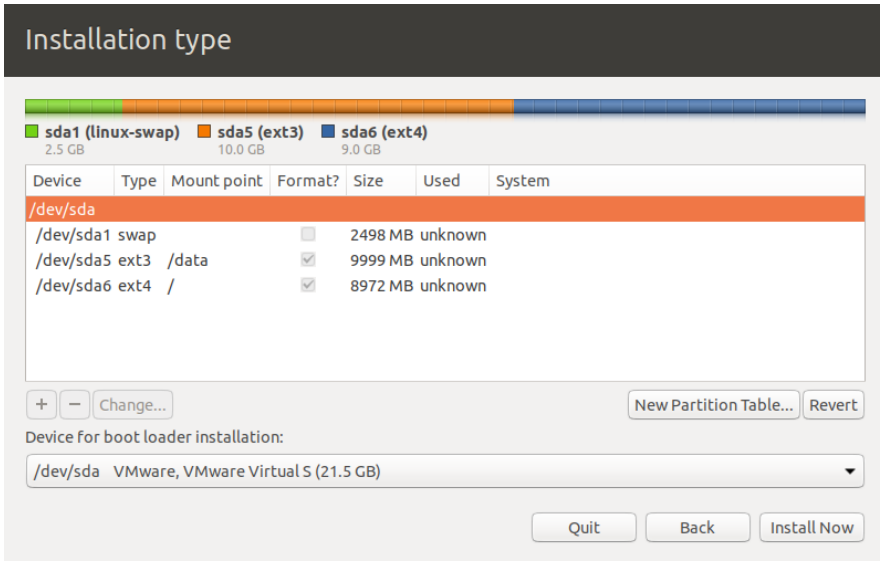
- I created a new virtual machine with the name of 'OSYS1000_VM03' and set the machine as showing in the below screenshot.



- I set the memory of the machine as 4 Gb.

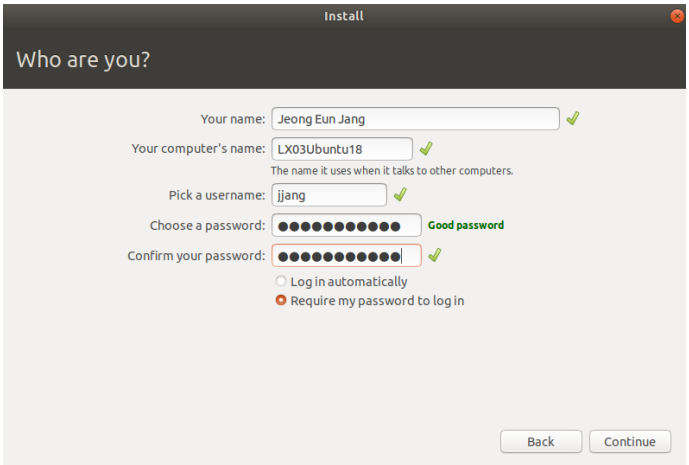


- I partitioned my virtual machine's hardware as following the instruction of the document such as 'swap 2.5 Gb' and 'data 10 Gb'.

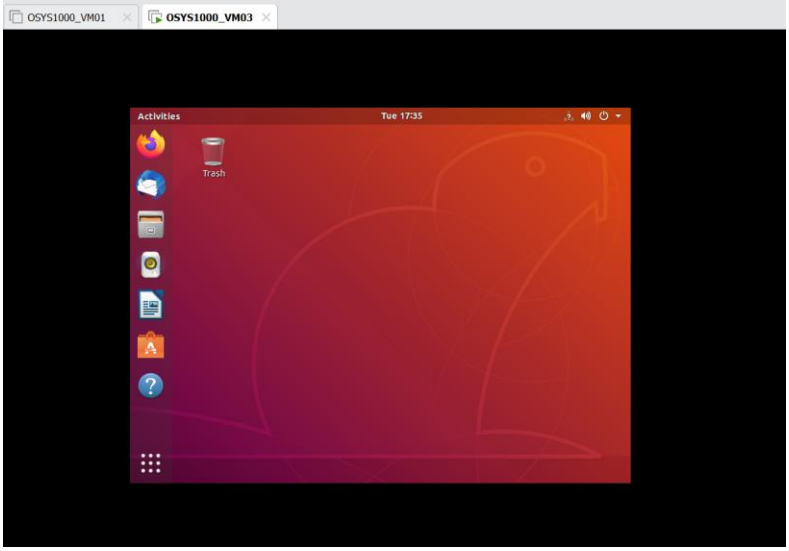


- I set the 'Language' as English/English(United States) and 'Date and Time' as Halifax.

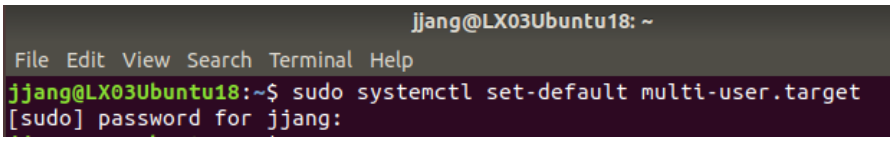
- I set my username as 'jjang' and the host name as 'LX03Ubuntu18' as following the naming convention.



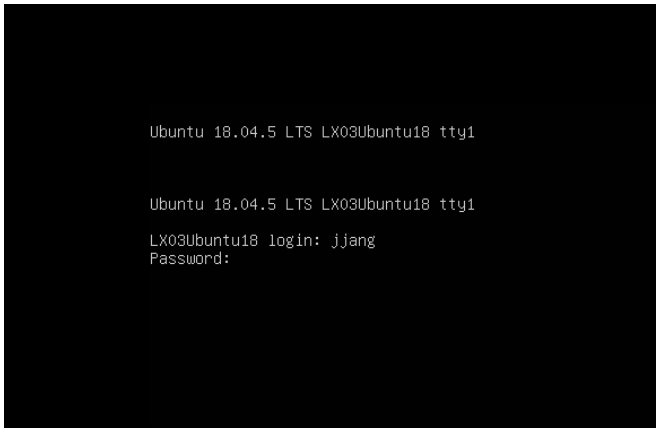
- Finally, I got my third Virtual Machine with Ubuntu18.04.5 LTS.



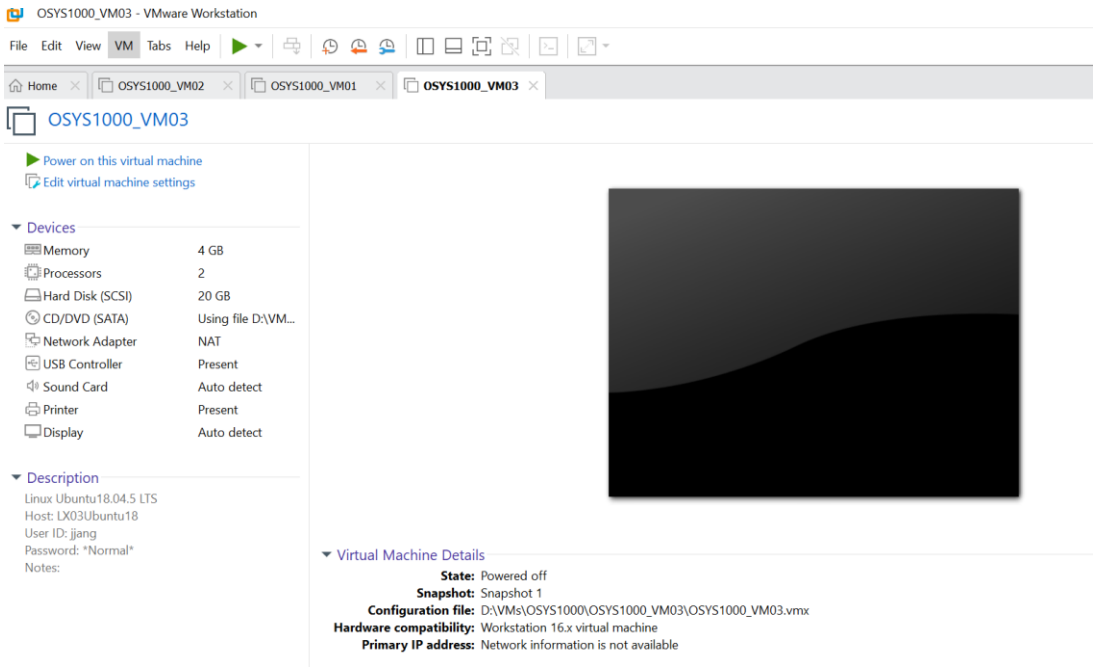
- To set up to boot into a non-graphical , multi-user run level, I used the syntax ‘sudo systemctl set-default multi-user.target’.



I restarted the machine, and I checked it, booted the machine with no GUI (Stack Exchange, 2019).



The screenshot below is my Virtual machine after I completed all set up.



2.3. Summary

: Ubuntu 18.04 is one of the highly recommended releases for Linux users. This version provides a long-term support, and it has a GNOME Shell desktop.

3. TASK 2: INSTALL OPENSSH, SAMBA, LAMP STACK

3.1. Objectives for this task are as follows:

: Install, properly figure out and demonstrate OpenSSH, Samba, and LAMP Stack in my new Linux environment.

3.2. OpenSSH

- Secure Shell (SSH) is a cryptographical network protocol used for a secure connection between a client and a server (Linuxize, 2019). Before installing SSH, I updated my system using a syntax 'sudo apt update' and check whether there is 'openSSH' in my system. I could see there was only 'openSSH-client' in my system.

```
jjang@LX03Ubuntu18:~$ dpkg -l | grep openssh
ii  openssh-client                          1:7.6p1-4ubuntu0.3
                amd64        secure shell (SSH) client, for secure access t
emote machines
jjang@LX03Ubuntu18:~$ sudo apt-get update
Hit:1 http://security.ubuntu.com/ubuntu bionic-security InRelease
Hit:2 http://ca.archive.ubuntu.com/ubuntu bionic InRelease
Hit:3 http://ca.archive.ubuntu.com/ubuntu bionic-updates InRelease
Hit:4 http://ca.archive.ubuntu.com/ubuntu bionic-backports InRelease
Reading package lists... Done
jjang@LX03Ubuntu18:~$ sudo apt-get install openssh-server
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
```

- So, I installed it using a command 'sudo atp-get install openssh-server' and checked the status.

```
jjang@LX03Ubuntu18:~$ dpkg -l | grep openssh
ii  openssh-client                          1:7.6p1-4ubuntu0.3
                amd64        secure shell (SSH) client, for secure access to r
emote machines
ii  openssh-server                          1:7.6p1-4ubuntu0.3
                amd64        secure shell (SSH) server, for secure access from
 remote machines
ii  openssh-sftp-server                     1:7.6p1-4ubuntu0.3
                amd64        secure shell (SSH) sftp server module, for SFTP a
ccess from remote machines
jjang@LX03Ubuntu18:~$
```

- After the installation completed, I verified the installation was successful and SSH service is running using a following command ‘sudo systemctl status ssh’. I checked the ‘Active: active (running).

```
jjang@LX03Ubuntu18:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: ena
   Active: active (running) since Wed 2021-04-07 12:54:00 ADT; 7min ago
 Main PID: 59881 (sshd)
    Tasks: 1 (limit: 4634)
   CGroup: /system.slice/ssh.service
           └─59881 /usr/sbin/sshd -D

Apr 07 12:54:00 LX03Ubuntu18 systemd[1]: Starting OpenBSD Secure Shell server..
Apr 07 12:54:00 LX03Ubuntu18 sshd[59881]: Server listening on 0.0.0.0 port 22.
Apr 07 12:54:00 LX03Ubuntu18 sshd[59881]: Server listening on :: port 22.
Apr 07 12:54:00 LX03Ubuntu18 systemd[1]: Started OpenBSD Secure Shell server.
lines 1-12/12 (END)
```

- Ubuntu comes with a firewall configuration tool called UFW. I used the command ‘sudo ufw allow ssh’ to completed setting SSH port.

```
jjang@LX03Ubuntu18:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
jjang@LX03Ubuntu18:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau
lt qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 00:0c:29:8c:0d:71 brd ff:ff:ff:ff:ff:ff
    inet 192.168.189.131/24 brd 192.168.189.255 scope global dynamic noprefixro
ute ens33
       valid_lft 1696sec preferred_lft 1696sec
    inet6 fe80::2035:7f46:2c0b:dde6/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
jjang@LX03Ubuntu18:~$
```

- To check my SSH connection over LAN, I found my ip address ‘192.168.189.131’. I was able to log in to remote machine by running the following ssh command (Linuxize, 2019):

  $ ssh username@ip address

```
jjang@LX03Ubuntu18:~$ ssh jjang@192.168.189.131
The authenticity of host '192.168.189.131 (192.168.189.131)' can't be establish
ed.
ECDSA key fingerprint is SHA256:1b0wE5j5PI1NANhhvt2AfrTOzm9RmNAOITXsYexupOQ.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.168.189.131' (ECDSA) to the list of known hosts
.
jjang@192.168.189.131's password:
```

- Now, I am logged in to my Ubuntu machine after entering the password.

```
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 5.4.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage
```

3.3. Samba

I followed all steps for the installation and configuration of Samba from the Linuxize article (Linuxize, 2019).

```
jjang@LX03Ubuntu18:~$ sudo apt-get install samba
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  attr ibverbs-providers libcephfs2 libibverbs1 libnl-route-3-200
  libpython-stdlib librados2 python python-crypto python-dnspython py
  python-minimal python-samba python-tdb python2.7 python2.7-minimal
  samba-common samba-common-bin samba-dsdb-modules samba-vfs-modules
  tdb-tools
```

I checked whether the Samba server is running well.

```
jjang@LX03Ubuntu18:~$ sudo systemctl status smbd
● smbd.service - Samba SMB Daemon
   Loaded: loaded (/lib/systemd/system/smbd.service; enabled; vendor preset: en
   Active: active (running) since Thu 2021-04-08 12:06:03 ADT; 2min 34s ago
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
 Main PID: 2824 (smbd)
   Status: "smbd: ready to serve connections..."
    Tasks: 4 (limit: 4633)
   CGroup: /system.slice/smbd.service
           ├─2824 /usr/sbin/smbd --foreground --no-process-group
           ├─2826 /usr/sbin/smbd --foreground --no-process-group
           ├─2827 /usr/sbin/smbd --foreground --no-process-group
           └─2829 /usr/sbin/smbd --foreground --no-process-group

Apr 08 12:06:03 LX03Ubuntu18 systemd[1]: Starting Samba SMB Daemon...
Apr 08 12:06:03 LX03Ubuntu18 systemd[1]: Started Samba SMB Daemon.
```

I can see that Samba has been installed and ready to be configured.

- Configuring firewall using ufw.

  The Ubuntu system is using ufw to manage firewall, I opened the ports by enabling the ‘Samba’ profile with this syntax:

  $sudo ufw allow ‘Samba’

```
jjang@LX03Ubuntu18:~$ sudo ufw allow 'Samba'
[sudo] password for jjang:
Rules updated
Rules updated (v6)
```

I forgot to take a screenshot, but I had created a backup of Samba before making changes to the Samba configuration file for future reference.

$sudo cp /etc/samba/smb.conf{,.backup}

I opened the default configuration file to make sure ‘server role’ was set to ‘standalone server’.

```
# Most people will want "standalone sever" or "member server".
# Running as "active directory domain controller" will require first
# running "samba-tool domain provision" to wipe databases and create a
# new domain.
   server role = standalone server
```

To restrict access to the Samba server only from my internal network, I remove the following two lines about interfaces and bind interfaces.

```
#### Networking ####

# The specific set of interfaces / networks to bind to
# This can be either the interface name or an IP address/netmask;
# interface names are normally preferred


# Only bind to the named interfaces and/or networks; you must use the
# 'interfaces' option above to use this.
# It is recommended that you enable this feature if your Samba machine is
# not protected by a firewall or is a firewall itself.  However, this
# option cannot handle dynamic or non-broadcast interfaces correctly.
```

I created a new user named ‘jeong’ for practice. I used this syntax:

$sudo useradd -M -d /samba/Jeong -s /usr/sbin/nologin -G sambashare

jeong

‘-M’ means ‘do not create the user’s home directory’.

‘d /samba/jeong’ means ‘set the user’s home directory to /samba/Jeong.

‘-s’ means ‘disable shell access for this user’.

‘-G’ means ‘add the user to the sambashare group’.

```
jjang@LX03Ubuntu18:~$ sudo useradd -M -d /samba/jeong -s /usr/sbin/nologin -G s
ambashare jeong
jjang@LX03Ubuntu18:~$ sudo mkdir /samba/jeong
jjang@LX03Ubuntu18:~$ sudo chown jeong:sambashare /samba/jeong
jjang@LX03Ubuntu18:~$ sudo chmod 2770 /samba/jeong
jjang@LX03Ubuntu18:~$ sudo smbpasswd -a jeong
New SMB password:
Retype new SMB password:
Added user jeong.
jjang@LX03Ubuntu18:~$ sudo smbpasswd -e jeong
Enabled user jeong.
```

I set the permissions to 2770 and the user ‘jeong’ could read and write to this file when the sadmin user creates a new file. I set the user password as well.

- Configuring Samba Shares.

I opened the Samba configuration file and appended the sections below.

```
[jjang]
comment = user account
path = /home/jjang
valid users = @sambashare @sadmin
read only = no
force create mode = 0660
force directory mode = 2770
browseable = yes
```

```
[jeong]
path = /samba/jeong
browseable = no
read only = no
force create mode = 0660
force directory mode = 2770
```

‘force directory mode’: Sets the permissions for the newly created directories in this share.

‘force create mode’: Sets the permissions for the newly created files in this share.

I restarted the Samba services with ‘sudo systemctl restart smbd’ and ‘sudo systemctl restart nmbd’.

I installed ‘smbclient’ as a tool to access Samba from the command line.

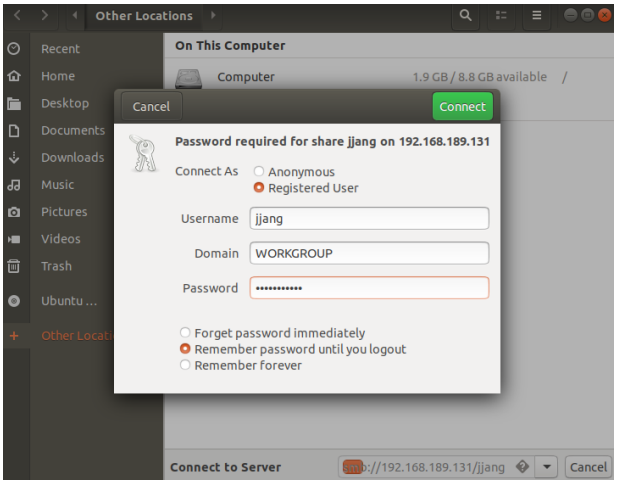
```
jjang@LX03Ubuntu18:~$ sudo apt install smbclient
[sudo] password for jjang:
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  cifs-utils heimdal-clients
```

- I installed the cifs-utils package for mounting the Samba share.

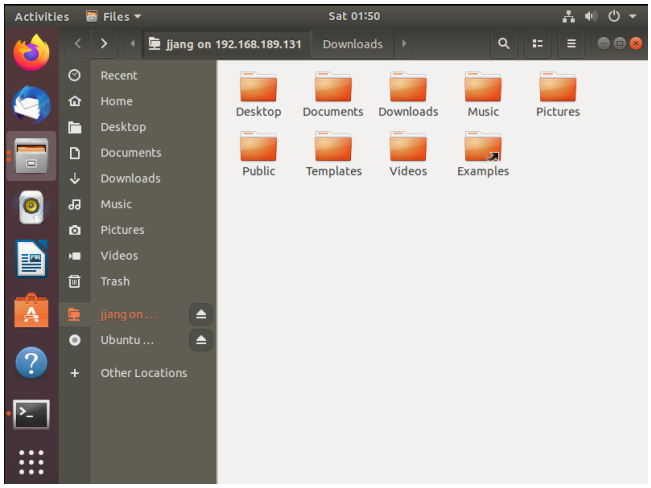
```
jjang@LX03Ubuntu18:~$ sudo apt install cifs-utils
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  keyutils winbind
```

```
jjang@LX03Ubuntu18:~$ sudo mkdir /mnt/smbmount
jjang@LX03Ubuntu18:~$ sudo mount -t cifs -o username=jjang //192.168.189.131/jj
ang /mnt/smbmount
Password for jjang@//192.168.189.131/jjang:  ************
```

The default file manager in Gnome has built-in option to access Samba shares.

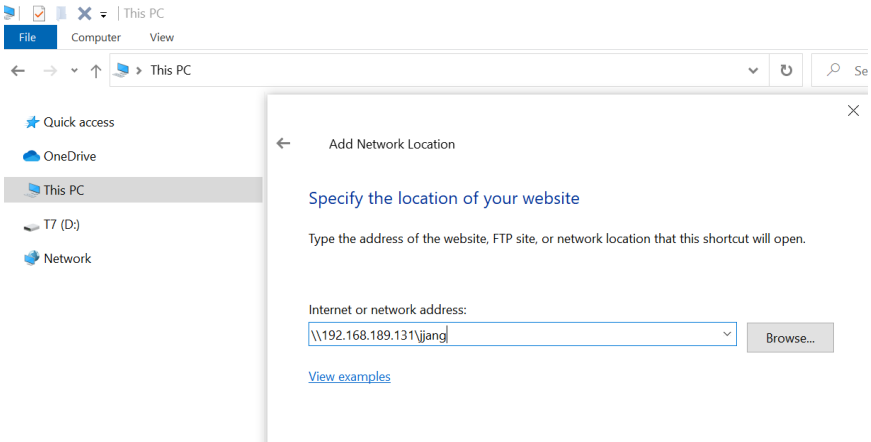


I entered the Samba share by clicking ‘Connect to Server’ with this format : smb://samba_hostname_or_server_ip/sharename. And then, I selected ‘Registered User’ and typed the username and password.

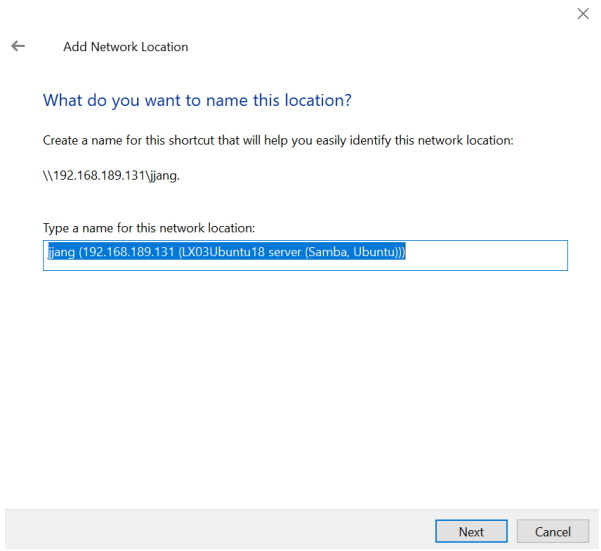


- Connecting to a Samba Share from Windows.

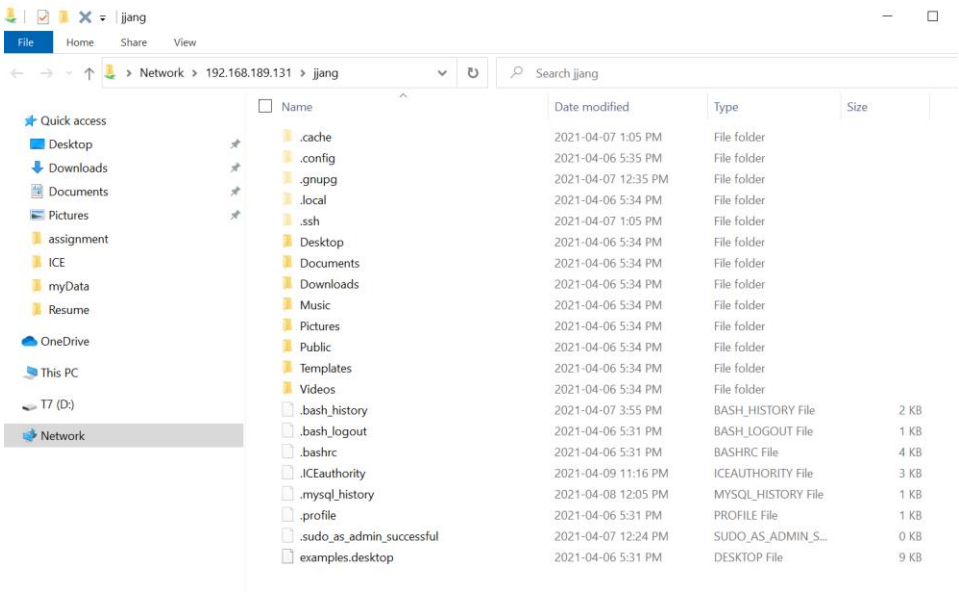
  I opened ‘File Explorer’ -> ‘This PC’ -> ‘Choose a custom network location’ and set my network address in the following format: \\samba_hostname_or_server_ip\sambaname



I finished the login credentials and then picked up my custom name for the network location as default one below.



After clicking 'Finish', the files on the Samba server were shown.



Finally, I checked the sharing was completed well.

```
jjang@LX03Ubuntu18:~$ pwd
/home/jjang
jjang@LX03Ubuntu18:~$ ls -al
total 96
drwxr-xr-x 15 jjang jjang 4096 Apr  9 23:16 .
drwxr-xr-x  3 root  root  4096 Apr  6 17:31 ..
-rw-------  1 jjang jjang 1906 Apr  7 15:55 .bash_history
-rw-r--r--  1 jjang jjang  220 Apr  6 17:31 .bash_logout
-rw-r--r--  1 jjang jjang 3771 Apr  6 17:31 .bashrc
drwx------ 11 jjang jjang 4096 Apr  7 13:05 .cache
drwx------ 11 jjang jjang 4096 Apr  6 17:35 .config
drwxr-xr-x  2 jjang jjang 4096 Apr  6 17:34 Desktop
drwxr-xr-x  2 jjang jjang 4096 Apr  6 17:34 Documents
drwxr-xr-x  2 jjang jjang 4096 Apr  6 17:34 Downloads
-rw-r--r--  1 jjang jjang 8980 Apr  6 17:31 examples.desktop
drwx------  3 jjang jjang 4096 Apr  7 12:35 .gnupg
-rw-------  1 jjang jjang 2394 Apr  9 23:16 .ICEauthority
drwx------  3 jjang jjang 4096 Apr  6 17:34 .local
drwxr-xr-x  2 jjang jjang 4096 Apr  6 17:34 Music
-rw-------  1 jjang jjang   18 Apr  8 12:05 .mysql_history
drwxr-xr-x  2 jjang jjang 4096 Apr  6 17:34 Pictures
-rw-r--r--  1 jjang jjang  807 Apr  6 17:31 .profile
drwxr-xr-x  2 jjang jjang 4096 Apr  6 17:34 Public
drwx------  2 jjang jjang 4096 Apr  7 13:05 .ssh
-rw-r--r--  1 jjang jjang    0 Apr  7 12:24 .sudo_as_admin_successful
drwxr-xr-x  2 jjang jjang 4096 Apr  6 17:34 Templates
drwxr-xr-x  2 jjang jjang 4096 Apr  6 17:34 Videos
```

3.4. LAMP Stack

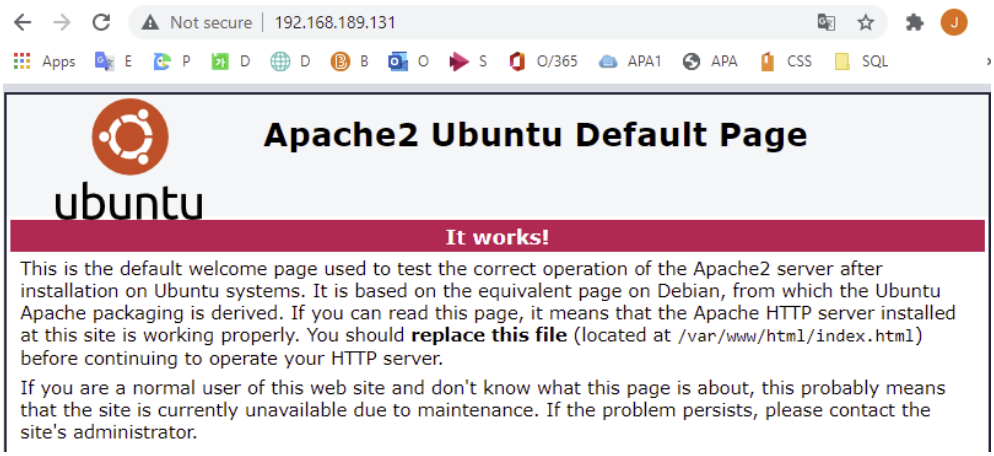
- A LAMP stack is a group of open-source software and the name is a combination of an acronym which represents the Linux operating system, the Apache web server, a MySQL database and PHP. They are typically installed together to experience a sever to host dynamic websites and web applications (Drake, 2018). At first, I checked the update and were ready to execute the operations with root privileges. I used the command ‘sudo apt install apache2’ to install Apache.

```
jjang@LX03Ubuntu18:~$ sudo apt update
[sudo] password for jjang:
Get:1 http://security.ubuntu.com/ubuntu bionic-security InRelease [88.7 kB]
Hit:2 http://ca.archive.ubuntu.com/ubuntu bionic InRelease
Get:3 http://ca.archive.ubuntu.com/ubuntu bionic-updates InRelease [88.7 kB]
Get:4 http://ca.archive.ubuntu.com/ubuntu bionic-backports InRelease [74.6 kB]
Get:5 http://ca.archive.ubuntu.com/ubuntu bionic-updates/main i386 Packages [1,
254 kB]
Get:6 http://ca.archive.ubuntu.com/ubuntu bionic-updates/main amd64 Packages [1
,979 kB]
Fetched 3,485 kB in 4s (809 kB/s)


Reading package lists... Done
Building dependency tree
Reading state information... Done
113 packages can be upgraded. Run 'apt list --upgradable' to see them.
jjang@LX03Ubuntu18:~$
jjang@LX03Ubuntu18:~$
jjang@LX03Ubuntu18:~$ sudo apt install apache2
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.2-0
```

- Next, I made the UFW firewall enabled to allow web traffic.

```
jjang@LX03Ubuntu18:~$ sudo ufw app list
Available applications:
  Apache
  Apache Full
  Apache Secure
  CUPS
  OpenSSH
jjang@LX03Ubuntu18:~$ sudo ufw app info "Apache Full"
Profile: Apache Full
Title: Web Server (HTTP,HTTPS)
Description: Apache v2 is the next generation of the omnipresent Apache web
server.

Ports:
  80,443/tcp
```

- Now, I can see my web server is correctly installed and accessible through my firewall as below screenshot.



- Installing MySQL

```
jjang@LX03Ubuntu18:~$ sudo apt install mysql-server
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libaio1 libevent-core-2.1-6 libhtml-template-perl mysql-client-5.7
  mysql-client-core-5.7 mysql-common mysql-server-5.7 mysql-server-core-5.7
```

I used ‘sudo apt install mysql-server’ command to install mysql in my server.

```
jjang@LX03Ubuntu18:~$ sudo mysql_secure_installation

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD PLUGIN can be used to test passwords
and improve security. It checks the strength of password
and allows the users to set only those passwords which are
secure enough. Would you like to setup VALIDATE PASSWORD plugin?
```

```
mysql> SELECT user,authentication_string,plugin,host FROM mysql.user;
+------------------+-------------------------------------------+-----------------------+-----------+
| user             | authentication_string                     | plugin                | host      |
+------------------+-------------------------------------------+-----------------------+-----------+
| root             |                                           | auth_socket           | localhost |
| mysql.session    | *THISISNOTAVALIDPASSWORDTHATCANBEUSEDHERE | mysql_native_password | localhost |
| mysql.sys        | *THISISNOTAVALIDPASSWORDTHATCANBEUSEDHERE | mysql_native_password | localhost |
| debian-sys-maint | *F25112C24D376A0FC055353373FC50B740C13C79 | mysql_native_password | localhost |
+------------------+-------------------------------------------+-----------------------+-----------+
4 rows in set (0.00 sec)
```

ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY 'password';

```
mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT user, authentication_string,plugin,host FROM mysql.user;
+------------------+-------------------------------------------+-----------------------+-----------+
| user             | authentication_string                     | plugin                | host      |
+------------------+-------------------------------------------+-----------------------+-----------+
| root             | *D17EFBBEBEE8E8A39AED92D697426861705E4157 | mysql_native_password | localhost |
| mysql.session    | *THISISNOTAVALIDPASSWORDTHATCANBEUSEDHERE | mysql_native_password | localhost |
| mysql.sys        | *THISISNOTAVALIDPASSWORDTHATCANBEUSEDHERE | mysql_native_password | localhost |
| debian-sys-maint | *F25112C24D376A0FC055353373FC50B740C13C79 | mysql_native_password | localhost |
| jjang            | *D17EFBBEBEE8E8A39AED92D697426861705E4157 | mysql_native_password | localhost |
+------------------+-------------------------------------------+-----------------------+-----------+
5 rows in set (0.00 sec)
```

I set the root MySQL user authenticates using a password. My database system was set up and I made a new database using MySQL.

```
mysql> create database db DEFAULT CHARACTER SET utf8;
Query OK, 1 row affected (0.00 sec)

mysql> create user jjang identified by [redacted]
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT ALL PRIVILEGES ON db.* TO 'jjang'@'localhost'
ea1206';
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| db                 |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.00 sec)
```

```
mysql> use db;
Database changed
mysql> create table Person(
    -> id bigint(20) unsigned not null auto_increment,
    -> name varchar(255) not null,
    -> address varchar(255) not null,
    -> primary key (id)
    -> ) DEFAULT CHARACTER SET utf8;
Query OK, 0 rows affected (0.17 sec)

mysql> show tables;
+--------------+
| Tables_in_db |
+--------------+
| Person       |
+--------------+
1 row in set (0.00 sec)

mysql> insert into Person(name, address) values('Jeong','Halifax');
Query OK, 1 row affected (0.00 sec)
```

I made a table named ‘Person’ and identified each attribute.

```
mysql> insert into Person(name, address) values('Jeong','Halifax');
Query OK, 1 row affected (0.00 sec)

mysql> select * from Person;
+----+-------+---------+
| id | name  | address |
+----+-------+---------+
|  1 | Jeong | Halifax |
+----+-------+---------+
1 row in set (0.00 sec)
```

I dropped the table and finished MySQL.

```
mysql> drop tables Person;
Query OK, 0 rows affected (0.00 sec)

mysql> show tables;
Empty set (0.00 sec)

mysql> exit
Bye
jjang@LX03Ubuntu18:~$
```

- PHP

  : Most web servers support sever-side scripting such as PHP. PHP help creating dynamic web pages that are not available in simple HTML documents or modify data from databases, etc.

  I installed PHP using a command ‘sudo apt install php libapche2-mod-php php-mysql’.

```
File  Edit  View  Search  Terminal  Help
jjang@LX03Ubuntu18:~$ sudo apt install php libapache2-mod-php php-mysql
[sudo] password for jjang:
Reading package lists... Done
Building dependency tree
Reading state information... Done
```

I modified the way that Apache served files to let it look for PHP files first when a directory was requested.

```
jjang@LX03Ubuntu18: ~
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3          /etc/apache2/mods-enabled/dir.conf          Modified

<IfModule mod_dir.c>
        DirectoryIndex index.php index.html index.cgi index.pl index.xhtml ind$
</IfModule>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet
```

I also checked the status of the apache2 using a ‘systemctl’ command.

```
jjang@LX03Ubuntu18:~$ sudo nano /etc/apache2/mods-enabled/dir.conf
jjang@LX03Ubuntu18:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset:
  Drop-In: /lib/systemd/system/apache2.service.d
           └─apache2-systemd.conf
   Active: active (running) since Wed 2021-04-07 15:30:38 ADT; 4min 24s ago
  Process: 9100 ExecStop=/usr/sbin/apachectl stop (code=exited, status=0/SUCCES
  Process: 9105 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCC
 Main PID: 9109 (apache2)
    Tasks: 6 (limit: 4633)
   CGroup: /system.slice/apache2.service
           ├─9109 /usr/sbin/apache2 -k start
           ├─9111 /usr/sbin/apache2 -k start
           ├─9112 /usr/sbin/apache2 -k start
           ├─9113 /usr/sbin/apache2 -k start
           ├─9114 /usr/sbin/apache2 -k start
           └─9115 /usr/sbin/apache2 -k start

Apr 07 15:30:38 LX03Ubuntu18 systemd[1]: Starting The Apache HTTP Server...
Apr 07 15:30:38 LX03Ubuntu18 apachectl[9105]: AH00558: apache2: Could not relia
Apr 07 15:30:38 LX03Ubuntu18 systemd[1]: Started The Apache HTTP Server.
lines 1-20/20 (END)
```

I tested that my system was configured properly for PHP by creating a basic PHP script called info.php.

```
jjang@LX03Ubuntu18: ~
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                    /var/www/html/info.ph

<?php phpinfo(); ?>
```

My test was successful, and I could visit the site below.

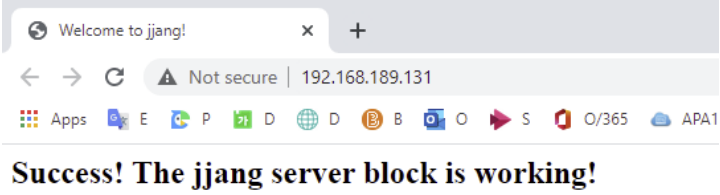
Not secure | 192.168.189.131/info.php

Apps E P D D B O S O/365 APA1 APA CSS SQL Linux Java etc.

PHP Version 7.2.24-0ubuntu0.18.04.7

System	Linux LX03Ubuntu18 5.4.0-42-generic #46~18.04.1-Ubuntu SMP Fri Jul 10 07:21:24 UTC 2020 x86_64
Build Date	Oct 7 2020 15:24:25
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/7.2/apache2
Loaded Configuration File	/etc/php/7.2/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/7.2/apache2/conf.d
Additional .ini files parsed	/etc/php/7.2/apache2/conf.d/10-mysqlnd.ini, /etc/php/7.2/apache2/conf.d/10-opcache.ini, /etc/php/7.2/apache2/conf.d/10-pdo.ini, /etc/php/7.2/apache2/conf.d/20-calendar.ini, /etc/php/7.2/apache2/conf.d/20-ctype.ini, /etc/php/7.2/apache2/conf.d/20-exif.ini, /etc/php/7.2/apache2/conf.d/20-fileinfo.ini, /etc/php/7.2/apache2/conf.d/20-ftp.ini, /etc/php/7.2/apache2/conf.d/20-gettext.ini, /etc/php/7.2/apache2/conf.d/20-iconv.ini, /etc/php/7.2/apache2/conf.d/20-json.ini, /etc/php/7.2/apache2/conf.d/20-mysqli.ini, /etc/php/7.2/apache2/conf.d/20-pdo_mysql.ini, /etc/php/7.2/apache2/conf.d/20-phar.ini, /etc/php/7.2/apache2/conf.d/20-posix.ini, /etc/php/7.2/apache2/conf.d/20-readline.ini, /etc/php/7.2/apache2/conf.d/20-shmop.ini, /etc/php/7.2/apache2/conf.d/20-sockets.ini, /etc/php/7.2/apache2/conf.d/20-sysvmsg.ini, /etc/php/7.2/apache2/conf.d/20-sysvsem.ini, /etc/php/7.2/apache2/conf.d/20-sysvshm.ini, /etc/php/7.2/apache2/conf.d/20-tokenizer.ini
PHP API	20170718
PHP Extension	20170718
Zend Extension	320170718
Zend Extension Build	API320170718,NTS
PHP Extension Build	API20170718,NTS

- Setting up Virtual Host

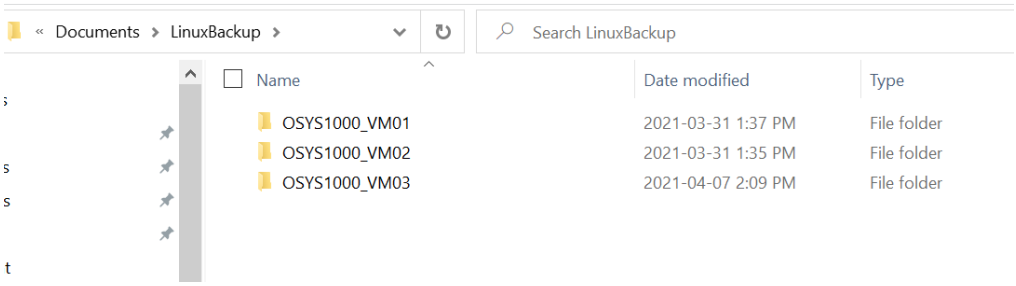


3.5. Summary

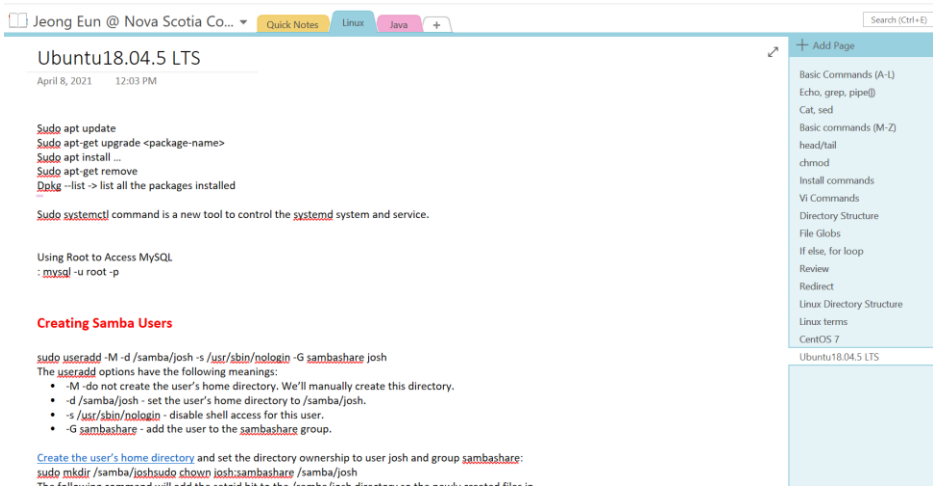
- I have learned how to install and enable SSH on my Ubuntu 18.04. It means that I can now login to my machine and perform common sysadmin tasks through the command prompt. In addition, SSH listens on port 22 as default.
- I have learned how to install Samba server on Ubuntu 18.04 and created a new user. I have connected the Samba server from Linux and Widnows.
- To help manage MySQL databases from web browser, I am going to set up PHPMyAdmin.

4. Toolbox & Gold copy

I made a gold copy for Ubuntu 18.04. in my 'LinuxBackup' folder. I encountered an error during installing mysql, but I was able to solve the problem using a gold copy. It was a valuable experience.



I have just added 'Ubuntu 18.04' page to save the different command from other Linux distribution and used commands for this assignment.



5. CONCLUSION

With this third Linux distribution, my official Linux project is over. It seems like yesterday that I was unfamiliar and struggling a little, starting my first Linux, Linux Mint. Now, it has already come to the last assignment. The name of 'Ubuntu' has the meaning of 'Humanity to others.' Just like its name, it focused on user convenience more than any other Linux distribution. Because of this attribute, I could finish my last assignment with no colossal problems. My Linux distributions installed on my Virtual Machine are the result that I have worked hard. I am sure that they will be valuable assets to me as a data administrator. During this summer vacation, I plan to challenge Debian with my fourth Linux distribution.

References

Drake M. (2018, April 27). How to Install Linux, Apache, MySQL, PHP (LAMP) stack On Ubuntu 18.04. https://www.digitalocean.com/community/tutorials/how-to-install-linux-apache-mysql-php-lamp-stack-ubuntu-18-04

Kumar, P. (2020, April 25). Ubuntu 18.04 LTS Desktop Installation Guide with Screenshots https://www.linuxtechi.com/ubuntu-18-04-lts-desktop-installation-guide-screenshots/

Linuxize. (2019, August 2). How to Enable SSH on Ubuntu 18.04 https://linuxize.com/post/how-to-enable-ssh-on-ubuntu-18-04/

Linuxize. (2019, January 27). How to Install and Configure Samba on Ubuntu 18.04. https://linuxize.com/post/how-to-install-and-configure-samba-on-ubuntu-18-04/

StackExchange. (2019). How to Disable GUI on Boot in 18.04 (Bionic Beaver)? https://askubuntu.com/questions/1056363/how-to-disable-gui-on-boot-in-18-04-bionic-beaver